# Interest-rate — iEarn by hand

Same coin (tUSDC), three lending venues, different supply APR. Move to the best — then **watch the crowd arb all three rates back together** (equilibrium). **Profit = spread × size × time.**

| IDLE WALLET | NET WORTH | CHASE P&L | YOUR BLENDED APR | TIME ELAPSED |
|---|---|---|---|---|
| 60,000 | 60,000 | +0 | 0.00% | 0 d |

▶ Run time speed 20 d/s ⚡ Move ALL to best venue

amount 20000 tUSDC

## • Aave-lite

**1.35%** supply APR

borrow APR 5.00% · util 30%

your supplied **0 tUSDC**

borrow demand (the market) 360,000

Supply | Withdraw all

## • Compound-lite

**7.20%** supply APR

borrow APR 10.00% · util 80%

your supplied **0 tUSDC**

borrow demand (the market) 800,000

Supply | Withdraw all

● **dYdX-lite** BEST

**19.12%** supply APR

borrow APR 25.00% · util 85%

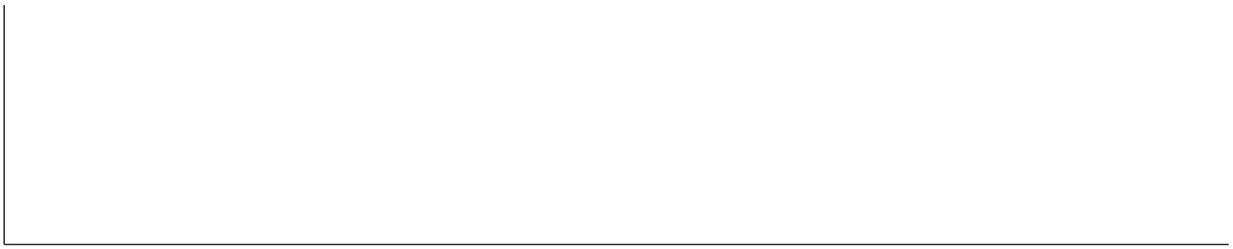
your supplied **0 tUSDC**

borrow demand (the market) 1,020,000

**Supply** **Withdraw all**

Three supply APRs over time — **the crowd arbitrages the gaps away** (normalization)

Aave-lite Compound-lite dYdX-lite spread (max−min) **17.77%**



Day **0**: best is **dYdX-lite @ 19.12%**. Your **60,000** sits idle at **0%** — deploy it. The crowd is arbing the three rates together — spread now **17.77%**. Chase vs sit: **+0**.

**The one idea (iEarn, 2020):** on-chain yields are **public readable state**, and moving your deposit is a **permissionless function**. So "put my stablecoin wherever it earns most" can be written as an unattended contract — no bank, no counterparty approval.

**Why the gap closes (convergence):** when you supply into the high-APR venue, its **utilization drops** (more cash, same borrowers) → its supply APR falls. Pile in enough and the three venues equalize. That is a market becoming efficient — just **slowly, over time**, unlike a swap arb that snaps shut in one trade.

**Re-open a gap:** drag a venue's **borrow-demand** slider up (a big borrower arrives) → its utilization and APR jump → chase again. This is what a rebalancing vault would do for you automatically — the next step.

MB/CSE 599 · AI Finance · interest-rate arbitrage (supply-side rate chasing) · same kink model as our LendingPool